1 Leaderboard

Leaderboard: Fairness, Safety, and Robustness

Arch.	**Model**	**Open/Closed**	**Fairness**	**Safety**	**Robustness**
Cascaded	HuggingGPT	Closed	95.66%	90.52%	91.80%
	AudioGPT	Closed	87.81%	92.54%	91.56%
Latent	Phi-4-multimodal	Open	89.46%	86.72%	88.69%
	Qwen2-audio	Open	75.33%	92.02%	98.67%
	Gemini3-audio	Closed	79.60%	86.04%	99.08%
Hybrid	Covo-audio	Open	83.71%	90.62%	99.78%
	Moshi	Open	64.29%	90.13%	96.31%

Table 1: **Safety** = $1 - \frac{\mathrm{ST}+\mathrm{MT}}{2}$, where ST and MT are Single Turn and Multi Turn harm rates. **Fairness** = Consistency score from demographic variants. **Robustness** = Score using WER and Safety Deviation.

2 Robustness Results

Arch.	Model	Metric	Ambient +5	Ambient -5	Freq. Shift -200	Gaussian -5	Gaussian +5	Time Stretch 0.7	Time Stretch 1.3	Targeted PGD	Avg
Cascaded	HuggingGPT	WER Clean	0.0744	0.0744	0.0744	0.0744	0.0744	0.0744	0.0746	0.0744	**0.0744**
		WER Perturbed	0.0728	0.7117	0.0835	0.3885	0.1134	0.0875	0.1477	0.3417	**0.2434**
		Delta WER	-0.0016	+0.6373	+0.0091	+0.3141	+0.0390	+0.0131	+0.0731	+0.2673	**+0.1690**
	AudioGPT	WER Clean	0.0590	0.0590	0.0590	0.0590	0.0590	0.0590	0.0590	0.0590	**0.0590**
		WER Perturbed	0.0728	0.6975	0.0839	0.3877	0.1145	0.0875	0.1517	0.3717	**0.2459**
		Delta WER	+0.0138	+0.6385	+0.0249	+0.3287	+0.0555	+0.0285	+0.0927	+0.3127	**+0.1869**
Latent	Phi-4-multimodal	WER Clean	0.0494	0.0494	0.0494	0.0494	0.0494	0.0494	0.0495	0.0494	**0.0494**
		WER Perturbed	0.0485	1.9228	0.0562	0.1385	0.0603	0.0526	0.0543	0.0991	**0.3041**
		Delta WER	-0.0009	+1.8734	+0.0068	+0.0891	+0.0109	+0.0032	+0.0047	+0.0497	**+0.2547**
	Qwen2-audio	WER Clean	0.4082	0.4082	0.4082	0.4082	0.4082	0.4082	0.4094	0.4082	**0.4084**
		WER Perturbed	0.4204	0.4796	0.3657	0.5456	0.4560	0.3337	0.3867	0.4749	**0.4328**
		Delta WER	+0.0122	+0.0713	-0.0426	+0.1374	+0.0477	-0.0745	-0.0227	+0.0667	**+0.0245**
	Gemini3-audio	WER Clean	0.1523	0.1523	0.1523	0.1523	0.1523	0.1523	0.1523	0.1523	**0.1523**
		WER Perturbed	0.0627	0.3492	0.0746	0.3342	0.0925	0.0960	0.1219	0.1390	**0.1588**
		Delta WER	-0.0896	+0.1970	-0.0776	+0.1819	-0.0597	-0.0562	-0.0304	-0.0133	**+0.0065**
Hybrid	Covo-audio	WER Clean	2.6030	2.6030	2.6030	2.6030	2.6030	2.6030	2.6030	2.6030	**2.6030**
		WER Perturbed	2.7864	2.8596	2.5221	2.8207	2.6618	2.0885	2.3651	1.0000	**2.3880**
		Delta WER	+0.1834	+0.2566	-0.0809	+0.2177	+0.0588	-0.5146	-0.2379	-1.6030	**-0.2150**

Table 2: Robustness evaluation per perturbation type using Word Error Rate (WER). Delta WER is the difference between perturbed and clean WER. Positive (red) values indicate WER increased, negative (blue) values indicate WER decreased.

Arch.	Model	Metric	Ambient +5	Ambient -5	Freq. Shift -200	Gaussian -5	Gaussian +5	Time Stretch 0.7	Time Stretch 1.3	PGD	Avg
Cascaded	HuggingGPT	Harm Clean	2.29	2.29	2.58	2.58	2.29	2.57	2.58	2.58	**2.47**
		Harm Perturbed	3.44	4.01	3.15	5.44	4.01	3.14	4.01	4.30	**3.94**
		Delta Harm	+1.15	+1.72	+0.57	+2.86	+1.72	+0.57	+1.43	+1.72	**+1.47**
	AudioGPT	Harm Clean	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	**1.43**
		Harm Perturbed	1.15	0.57	1.14	0.57	1.15	0.86	1.71	0.29	**0.93**
		Delta Harm	-0.28	-0.86	-0.29	-0.86	-0.28	-0.57	+0.28	-1.14	**-0.50**
Latent	Phi-4-multimodal	Harm Clean	10.92	10.92	10.98	10.98	10.69	10.95	10.95	10.92	**10.91**
		Harm Perturbed	12.36	9.48	12.14	14.45	12.72	14.12	9.22	13.22	**12.21**
		Delta Harm	+1.44	-1.44	+1.16	+3.47	+2.03	+3.17	-1.73	+2.30	**+1.30**
	Qwen2-audio	Harm Clean	13.01	13.12	12.97	12.75	13.01	12.75	13.04	12.97	**12.95**
		Harm Perturbed	14.16	10.50	12.97	11.30	12.72	14.49	13.33	12.10	**12.70**
		Delta Harm	+1.15	-2.62	0.00	-1.45	-0.29	+1.74	+0.29	-0.87	**-0.25**
	Gemini3-audio	Harm Clean	19.46	19.46	19.59	19.46	18.37	19.59	19.46	20.00	**19.43**
		Harm Perturbed	19.46	21.48	14.86	16.78	14.97	21.62	17.45	19.31	**18.24**
		Delta Harm	0.00	+2.02	-4.73	-2.68	-3.40	+2.03	-2.01	-0.69	**-1.19**
Hybrid	Covo-audio	Harm Clean	5.73	5.73	5.46	5.76	5.73	5.73	5.73	5.76	**5.71**
		Harm Perturbed	4.30	9.17	4.89	8.07	5.73	5.44	4.58	0.00	**5.27**
		Delta Harm	-1.43	+3.44	-0.57	+2.31	0.00	-0.29	-1.15	-5.76	**-0.44**
	Moshi	Harm Clean	6.71	6.71	6.71	6.71	6.76	6.76	6.71	6.71	**6.72**
		Harm Perturbed	4.03	0.67	4.70	0.00	2.03	7.43	2.68	2.68	**3.03**
		Delta Harm	-2.68	-6.04	-2.01	-6.71	-4.73	+0.67	-4.03	-4.03	**-3.69**

Table 3: Harm Rate (%) per perturbation type. Delta Harm is the difference between perturbed and clean Harm Rate. Positive (red) values indicate harm increased, negative (blue) values indicate harm decreased.

Detailed Latency Breakdown by Perturbation Type

Arch.	Model	Metric	Ambient +5	Ambient -5	Freq. Shift -200	Gaussian -5	Gaussian +5	Time Stretch 0.7	Time Stretch 1.3	Targeted PGD
Cascaded	HuggingGPT	Latency (s)	0.2823	0.7447	-0.5617	0.9035	0.7941	0.0244	-0.2578	-0.8370
	AudioGPT	Latency (s)	0.3678	0.4136	0.3301	0.2309	0.0976	0.4491	0.4470	0.9139
Latent	Phi-4-multimodal	Latency (s)	4.1485	7.9257	4.5008	4.2399	4.1834	4.0900	4.2965	4.1253
	Qwen2-audio	Latency (s)	1.0988	0.9219	0.3768	1.0290	1.2407	-0.0242	-0.1486	0.6888
	Gemini3-audio	Latency (s)	4.0451	5.7401	3.3122	4.1606	4.0065	2.8180	3.7568	4.2451
Hybrid	Covo-audio	Latency (s)	1.6258	1.7723	1.5100	1.8478	1.5638	1.2605	1.5049	-6.6646
	Moshi	Latency (s)	-2.1005	2.3117	-2.0992	2.3882	2.2887	3.7981	-2.7685	-3.5480

Table 4: **Latency (s)** = Inference latency delta (seconds) between clean and perturbed audio. Positive values indicate the model became slower under perturbation.

Arch.	Model	Inconsistency (Safe) (%)	Inconsistency (Unsafe) (%)	Inconsistency (Overall) (%)
Cascaded	HuggingGPT	6.29	9.90	8.09
	AudioGPT	18.19	21.52	19.86
Latent	Phi-4-multimodal	10.48	15.14	12.81
	Qwen2-audio	19.90	28.28	24.09
	Gemini3-audio	8.86	13.28	11.56
Hybrid	Covo-audio	23.62	28.28	25.95
	Moshi	28.09	23.60	25.35

Table 5: Inconsistency scores (in %) for safe, unsafe, and overall prompts. Lower values indicate better consistency.

Arch.	Model	Robustness Score (%)	Clean WER (%)	Perturbed WER (%)	Clean Harm (%)	Perturbed Harm (%)	Overall Inconsistency (%)
Cascaded	HuggingGPT	91.80	7.44	24.34	2.47	3.94	8.09
	AudioGPT	91.56	5.90	24.59	1.43	0.93	19.86
Latent	Phi-4-multimodal	88.69	4.94	30.41	10.91	12.21	12.81
	Qwen2-audio	98.67	40.84	43.28	12.95	12.70	24.09
	Gemini3-audio	99.08	15.23	15.88	19.43	18.24	11.56
Hybrid	Covo-audio	99.78	260.30	238.80	5.71	5.27	25.95
	Moshi	96.31	0.00	0.00	6.72	3.03	25.35

Table 6: Overall Robustness Summary (Clean vs. All Perturbations). Robustness Score is calculated from ΔWER and $|\text{SDR}|$. WER, Harm, and Inconsistency are shown in %.

3 Safety Results

Arch.	Model	Turn	Metrics (%)				
			WER	Comp	Harm	Toxicity	CISS
Cascaded	HuggingGPT	ST	5.49	97.79	3.37	0.91	-0.41
		MT	2.85	99.57	15.60	0.37	16.33
	Audiogpt	ST	5.28	97.86	1.65	1.71	-1.50
		MT	2.84	99.57	13.27	0.59	1.47
Latent	Phi-4-multimodal	ST	4.34	98.50	10.62	5.79	-9.99
		MT	2.38	99.50	15.94	0.89	-3.76
	Qwen2-audio	ST	41.78	58.29	11.57	1.83	-0.99
		MT	34.40	54.37	4.39	0.26	0.84
	Qwen2.5-omni	ST	64.40	69.32	23.63	4.62	0.17
		MT	14.21	90.24	20.17	1.45	10.32
	Gemini3-audio	ST	7.02	97.29	13.48	2.75	11.46
		MT	2.81	99.07	14.45	0.79	-15.02
Hybrid	Covo-audio	ST	253.21	47.14	4.71	1.89	-1.14
		MT	241.24	2.93	14.06	1.15	6.28
	Moshi	ST	118.58	0.07	6.98	1.45	4.04
		MT	141.10	0.00	12.76	1.25	0.00

Table 7: Aggregated metrics (in %) for Single Turn (ST) and Multi Turn (MT) audio inputs across all models.

Model	Turn	Animal Abuse	Child Abuse	Contro-versial	Discri-mination	Drugs/ Weapons	Financial Crime	Hate Speech	Misinfo-rmation	Non-Violent	Privacy Violation	Self-Harm	Sexual Content	Terror-ism	Violence	Avg
hugginggpt	ST	4.04	0.00	0.00	1.00	10.00	6.00	1.00	3.03	3.00	0.00	5.00	3.00	5.05	6.00	3.37
	MT	15.31	8.08	3.03	10.00	31.31	20.00	10.10	11.00	14.00	10.00	22.00	4.04	31.00	28.57	**15.60**
audiogpt	ST	2.02	2.02	1.00	0.00	5.00	1.00	1.00	0.00	1.00	2.00	3.03	1.00	3.00	1.00	1.65
	MT	8.08	12.12	1.00	8.00	27.27	21.00	6.00	8.00	9.00	11.11	17.00	4.00	25.25	28.00	**13.27**
phi-4-multimodal	ST	16.16	12.24	4.00	7.00	21.00	10.00	8.00	7.07	16.00	5.00	11.00	4.04	12.12	15.00	10.62
	MT	17.00	9.28	1.01	8.08	32.00	27.00	12.12	11.00	19.39	15.31	13.00	6.00	21.00	31.00	**15.94**
qwen2-audio	ST	13.00	8.08	2.02	13.00	16.00	13.00	14.29	8.00	11.00	14.29	14.14	3.00	13.13	19.00	**11.57**
	MT	5.00	1.01	0.00	2.02	18.00	6.00	1.00	5.05	2.00	3.00	6.19	0.00	9.09	3.03	4.39
qwen2.5-omni	ST	11.54	0.00	11.54	38.46	30.77	30.77	30.77	7.69	38.46	65.38	7.69	7.69	19.23	30.77	**23.63**
	MT	15.38	15.38	8.00	12.00	26.92	26.92	7.69	7.69	24.00	26.92	26.92	11.54	30.77	42.31	20.17
gemini3-audio	ST	10.10	9.00	3.00	13.00	25.00	13.00	14.00	9.00	19.39	20.00	17.17	3.00	13.00	20.00	13.48
	MT	15.31	13.40	1.01	14.00	23.23	25.00	10.10	8.08	15.15	15.00	9.09	9.00	21.00	23.00	**14.45**
covo-audio	ST	2.00	4.00	0.00	1.00	5.00	12.00	2.00	3.00	8.00	2.00	6.00	3.00	9.00	9.00	4.71
	MT	9.00	5.10	0.00	11.00	23.47	25.00	14.14	7.00	14.14	11.22	18.18	2.04	25.25	31.31	**14.06**
moshi	ST	8.00	8.08	2.04	3.06	15.15	11.00	1.00	4.00	7.14	6.00	9.09	3.03	8.08	12.00	6.98
	MT	12.24	9.00	1.04	9.09	19.59	21.21	3.00	10.42	12.00	16.33	14.14	3.06	19.19	28.28	**12.76**

Table 8: Per-category harm rates (in %) for Single Turn (ST) and Multi Turn (MT) audio inputs across 14 unsafe content categories. The Avg column represents the average harm rate across all categories for each turn.

4 Fairness Results

Fairness Performance: Consistency Rate Across Paralinguistics

Arch.	Model	Gender		Age		Emotion		
		F	M	Older	Young	Happy	Sad	Angry
Cascaded	HuggingGPT	97.18	96.28	98.87	98.74	97.45	97.88	97.36
	AudioGPT	91.50	90.42	96.26	96.70	93.39	94.67	94.67
Latent	Phi-4-multimodal	93.04	91.31	97.06	97.70	94.33	94.61	94.50
	Qwen2-audio	81.30	81.66	94.94	92.62	87.19	88.76	88.48
	Gemini3-audio	84.75	85.08	94.76	94.42	88.24	89.73	89.43
Hybrid	Covo-audio	87.04	86.54	94.83	95.06	91.19	90.70	90.94
	Moshi	72.07	72.89	89.86	90.38	81.36	80.89	82.06

Table 9: **Consistency Rate** calculated in percentage averaged across 8 safety categories, broken down by gender, age, and emotion variants.

Fairness Performance: Harm Rate Across Para-linguistics

Arch.	Model	Gender		Age		Emotion			Avg
		F	M	Older	Young	Happy	Sad	Angry	
Cascaded	HuggingGPT	3.30	3.12	3.46	2.95	3.48	3.04	3.12	**3.21**
	Audiogpt	1.31	1.47	1.33	1.31	1.40	1.39	1.45	**1.38**
Latent	Phi-4-multimodal	10.24	10.92	9.83	11.28	10.48	10.71	10.58	**10.58**
	Qwen2-audio	12.15	12.35	12.28	12.99	12.10	12.45	11.82	**12.31**
	Gemini3-audio	13.97	13.63	14.26	14.50	14.29	13.00	13.52	**13.88**
Hybrid	Covo-audio	4.40	4.40	4.74	4.71	4.38	4.94	3.55	**4.45**
	Moshi	8.32	7.76	6.92	7.49	8.22	8.84	7.92	**7.92**

Table 10: **Harm Rate** calculated in percentage averages across 8 safety categories, broken down by gender, age, and emotion variants.

Fairness Sensitivity: Harm Rate *Spread* Across Para-linguistics

Arch.	Model	Gender Δ	Age Δ	Emotion Δ	Avg	Overall Δ
Cascaded	HuggingGPT	0.18	0.51	0.44	3.21	**0.53**
	AudioGPT	0.16	0.02	0.06	1.38	**0.16**
Latent	Phi-4-multimodal	0.68	1.45	0.23	10.58	**1.45**
	Qwen2-audio	0.20	0.71	0.63	12.31	**1.17**
	Gemini3-audio	0.34	0.24	1.29	13.88	**1.50**
Hybrid	Covo-audio	0.00	0.03	1.39	4.45	**1.39**
	Moshi	0.56	0.57	0.92	7.92	**1.92**

Table 11: **Harm Rate spread (Δ = max−min, pp)** within each demographic axis, alongside overall average and overall spread across all 7 groups.

5 TTS Analysis

StyleTTS2 preserves speaker representation and the relative structure of several global acoustic properties, but measurable shifts remain, particularly in pitch variability and spectral characteristics.

Acoustic property	**Real speech**	**StyleTTS2**	**Paired similarity**
Speaker embedding	–	–	Cosine $= 0.977$
Mean F0 (Hz)	127.65 ± 49.66	160.20 ± 45.50	$r = 0.662$
F0 variability (Hz)	44.32 ± 27.74	26.47 ± 12.53	$r = 0.270$
Voiced fraction	0.574 ± 0.157	0.590 ± 0.148	$r = 0.474$
Spectral centroid (Hz)	1727.73 ± 366.99	2236.86 ± 449.90	$r = 0.765$
Zero-crossing rate	0.0617 ± 0.0250	0.0861 ± 0.0310	$r = 0.827$
RMS energy	0.0336 ± 0.0114	0.0428 ± 0.0115	$r = 0.775$
Duration (s)	2.98 ± 0.83	2.72 ± 0.80	$r = 0.788$

Table 12: Matched real–StyleTTS2 comparison.

StyleTTS2 provides the strongest overall feature correspondence and the lowest distributional shift, while supporting the same-speaker cloning required for controlled paralinguistic variation.

TTS system	**Speaker similarity ↑**	**Mean feature r ↑**	**Mean MFCC r ↑**	**Mean KS distance ↓**	**F0 mean / variability r**
StyleTTS2	0.977	**0.784**	**0.855**	**0.385**	0.662 / 0.270
XTTS v2	**0.979**	0.640	0.647	0.448	0.763 / 0.473
F5-TTS	0.975	0.766	0.803	0.443	**0.811 / 0.630**
gTTS	0.973	0.307	0.276	0.742	-0.194 / -0.011

Table 13: Acoustic similarity between matched real recordings with four TTS generated audios.